

```
# Subjects must match exactly, or both are CAT
# with matching CAT ID and acceptable CAT version
return (acl_subject == isd_subject) or
    (is_cat(acl_subject) and is_cat(isd_subject) and
    (get_cat_id(acl_subject) == get_cat_id(isd_subject)) and
    (get_cat_version(isd_subject) >= get_cat_version(acl_subject)))

def add_granted_privilege(granted_privileges, privilege):
    # Add the new privilege to the granted privileges set
    granted_privileges.add(privilege)
    # Also add any privileges subsumed by the new privilege
    if (privilege == PrivilegeEnum.Operate):
        granted_privileges.add(PrivilegeEnum.View)
    elif (privilege == PrivilegeEnum.Manage):
        granted_privileges.add(PrivilegeEnum.Operate)
        granted_privileges.add(PrivilegeEnum.View)
    elif (privilege == PrivilegeEnum.Administer):
        granted_privileges.add(PrivilegeEnum.Manage)
        granted_privileges.add(PrivilegeEnum.Operate)
        granted_privileges.add(PrivilegeEnum.View)

def get_granted_privileges(acl, subject_desc, endpoint_id, cluster_id) ->
set[Privilege]:
    # Granted privileges set is initially empty
    granted_privileges = set()

    # PASE commissioning channel implicitly grants administer privilege to commissioner
    if subject_desc.AuthMode == AuthModeEnum.PASE and
subject_desc.IsCommissioneeDuringCommissioning():
        add_granted_privilege(granted_privileges, PrivilegeEnum.Administer)

    for acl_entry in acl:
        # End checking if highest privilege is granted
        if PrivilegeEnum.Administer in granted_privileges: break

        # FabricIndex must match, there are no valid entries with FabricIndex == 0
        # other than the implicit PASE entry, which we will not see explicitly in the
        # access control list
        if acl_entry.FabricIndex == 0: continue
        if acl_entry.FabricIndex != subject_desc.FabricIndex: continue

        # Auth mode must match
        if acl_entry.AuthMode != subject_desc.AuthMode: continue
```